of the SROs who charge the fees.[2469] The Commission recognizes the risk that Rule 614(d)(9) and the extension of Regulation SCI to include competing consolidators could lead to less competition in the competing consolidator market,[2470] which could reduce the incentives of competing consolidators to reduce the cost and improve the speed and quality of consolidated market data. However, the Commission believes that the risk that there will be insufficient competition among competing consolidators is low.[2471] To the extent there is not sufficient competition among competing consolidators, it could make consolidated market data less of a viable substitute for proprietary data feeds, which would reduce the competitive pressures consolidated market data would impose on proprietary data feeds.

Third, the Commission expects the new decentralized consolidation model for consolidated market data to create competitors to market data aggregators for two reasons. First, the potential revenues from becoming a competing consolidator may cause new firms to enter the market for the consolidation and distribution of market data. Second, some market participants who currently use market data aggregators that do not choose to become competing consolidators may switch to getting consolidated market data products from a competing consolidator. This could have two effects: the competition could lead to lower prices and higher quality in the market data aggregator business, but it could also lead to fewer market data aggregators if the competition from the consolidated market data system makes it no longer

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2469 See supra Section V.C.4(a).

2470 For discussion of Regulation SCI requirements on competition, see supra Section V.C.2(a)(i)a.

2471 See supra Section V.C.2(a)(ii).

viable for some market data aggregators to offer their services to market participants who still wish to use proprietary data feeds.[2472] The latter could lead to higher prices in the market data aggregator space.[2473] In addition, some of these market data aggregators may choose to become competing consolidators, which could have two effects: it could cause market data aggregators to leave the proprietary feed aggregation space thereby reducing the competition in that space, or it could cause market data aggregators to use the economies of scale and the additional profits they may derive from being a competing consolidator to improve their offerings as a market data aggregator of proprietary feeds. Depending on which effect dominates, competition in the market data aggregator space could increase or decrease, which in turn could lead to lower or higher prices, respectively. The Commission recognizes that subjecting competing consolidators that fall above the market data revenue threshold to the requirements of Regulation SCI could diminish the ability of market data aggregators who become SCI competing consolidators to compete in the market data aggregator space.[2474] If a market data aggregator becomes an SCI competing consolidator, the requirements of being an SCI entity could also extend to their

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[2472] The Commission acknowledges that fewer competitors could decrease or increase efficiency in the market data aggregator business. On the one hand, fewer competitors could reduce the incentives for market data aggregators to innovate, which could reduce efficiency. On the other hand, fewer competitors could also improve efficiency if the firms that exited the market did not aggregate market data as efficiently as the firms that remained.

[2473] As discussed above, consolidated market data would not be a perfect substitute for proprietary data feeds, so there would still be demand for proprietary data. Since not all firms' aggregate proprietary data themselves, there would still be a demand for third-party aggregators to perform this function.

[2474] See supra Section III.F.

aggregation of proprietary market data.[2475] These requirements could raise their costs, which could reduce their ability to compete with other market data aggregators that are not competing consolidators.

Fourth, the Commission expects that the expanded content and reduced latency of core market data provided by this final rule may increase competition in the broker-dealer business by improving the ability of some broker-dealers who currently access core data to execute orders.[2476] It is the Commission's understanding that some broker-dealers that do not subscribe to all of the current proprietary DOB feeds rely solely on the exclusive SIPs today and that this makes them uncompetitive in the market for offering execution services to the most transaction-cost-sensitive market participants. The new decentralized consolidation model with expanded core data will reduce the latency and expand the information delivered to broker-dealers who subscribe to core data, possibly without raising data prices. This in turn would allow broker-dealers that subscribe to consolidated data to improve their order execution services and compete more effectively with broker-dealers who subscribe to proprietary DOB feeds. This will lead to greater competition between broker-dealers, which could benefit investors by resulting in lower prices for and higher quality of broker-dealer execution services.[2477]

Fifth, the Commission believes that the final rule could affect competition between exchanges. As discussed above, the final enhancements to core data could increase competition between consolidated market data and proprietary data feeds, which could lead to exchanges

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2475 See supra Section V.C.2(e)(ii).

2476 See supra Section V.C.4(b).

2477 See supra Sections V.B.3(e), V.C.4(b).

charging lower fees for proprietary market data.[2478] If these lower fees do not result in more subscribers to proprietary market data, it would lead to a decline in revenues from proprietary market data for SROs.[2479] Additionally, the amendments could affect competition in the market for exchange data connectivity. If some current subscribers to proprietary market data decide to only receive consolidated market data products from competing consolidators, they could also reduce the exchange connectivity services that they currently use. In turn, this could reduce the revenue that some exchanges earn from connectivity services. Additionally, new connectivity fees may be proposed for core data use cases, which could potentially increase or decrease the revenue exchanges earn from connectivity.[2480] It is the Commission's understanding that revenues from proprietary market data and connectivity services are a substantial portion of overall revenues for many exchanges.[2481] It is also the case that changes to the fees set by the effective national market system plan(s) for consolidated market data may result in lower revenues redistributed back to the exchanges, further contributing to a loss of revenue. It is possible that an exchange group could close some or all of its exchanges if the revenues from consolidated market data did not increase and revenues from proprietary market data and connectivity services were to decline to a level that a given exchange or exchange group is no

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2478 See supra Section V.C.4(a). One commenter stated that it was "not clear how such competition could occur, given that the Proposal is to authorize the NMS Plan to set all fees, including fees for proprietary data products, which contain core data." See Nasdaq Letter IV at 48. The amendments do not authorize the effective national market system plan(s) to set fees for proprietary data products, but instead for the data content underlying consolidated market data.

2479 In addition to adjusting fees, SROs could also redesign their proprietary market data product lines to try and increase revenue. However, it is possible that demand for these new products would not be sufficient to offset the decline in revenues from proprietary market data.

2480 See supra Section V.C.4(a).

2481 See supra Section V.B.3(b).

longer able to cover operating expenses. The Commission is unable to quantify the likelihood that an exchange will cease operating because it would depend on the fees and revenue allocation for consolidated market data. However, the Commission believes that it is unlikely exchanges will be forced to leave the market.

Even if an exchange were to exit, the Commission does not believe this would significantly impact competition in the market for trading services because the market is served by multiple competitors, including off-exchange trading venues. Consequently, if an exchange were to exit the market, demand is likely to be swiftly met by existing competitors. The Commission recognizes that small exchanges may have unique business models that are not currently offered by competitors, but the Commission believes a competitor could create similar business models if demand were adequate, and if they did not do so, it seems likely new entrants would do so if demand were sufficient.

One commenter stated that exchanges might be forced to increase fees for trading services in order to offset losses that might result from changes to core data fees.[2482] The Commission does not believe that the final amendments are likely to result in an increase in trading service fees, because losing revenue does not necessarily make it optimal for a firm to increase its fees. The Commission has discussed this point in the context of lost revenue specifically in proprietary data fees above, [2483] and believes that the same logic applies to the case of lost revenue from changes to core data fees as well.

2482 See, e.g., Nasdaq Letter IV at 4.

2483 See supra Section V.C.4(a) for additional discussion of the potential for trading fees to increase.

A commenter stated that the Commission did not consider the impact of "changes to market data fees" on SRO funding.[2484] The Commission acknowledges that if NMS data plan fees change such that revenue to SROs decline, then this could be an additional source of revenue loss to SROs from this rule. This would be in addition to the loss in proprietary data and connectivity discussed here, and the Commission believes the above discussion of the consequences of such losses, which was included in the Proposing Release,[2485] adequately analyzes the potential effects of SROs losing revenue, including from effective national market system plan data revenue. Furthermore, in response to this commenter's concern that the Commission does not recognize that there will be a "reduction in funding from proprietary feeds,"[2486] this discussion of the effect of the loss of proprietary data revenue above, which was included in the Proposing Release,[2487] analyzes such possibilities and their effects.

A commenter stated that the Commission failed to consider the possibility that SROs would be unable to perform their regulatory responsibilities if they were to lose revenue as a result of these final amendments.[2488] The Commission believes that this possibility is covered in the above discussion, which was included in the Proposing Release,[2489] through the discussion of the potential for exchanges to exit.

2484 See NYSE Letter II at 22.

2485 See Proposing Release, 85 FR at 16860.

2486 See NYSE Letter II ant 22

2487 See Proposing Release, 85 FR at 16860.

2488 See Nasdaq Letter IV at 50.

2489 See Proposing Release, 85 FR at 16860.

Sixth, the Commission believes that the final rule will affect competition between traders.[2490] The Commission believes that traders will be affected differently based on the type of market data they use when making trading decisions. For the purposes of this discussion, traders who subscribe to different types of market data can broadly be grouped into three categories: (1) traders who use proprietary DOB feeds received directly from the SROs and self-aggregate, (2) traders who use market data aggregators to aggregate proprietary DOB feeds, and (3) traders who use core data (currently from the exclusive SIPs and, under the final rule, competing consolidators).[2491] The Commission believes that under the final rule the core data would be of higher quality, and thus the value to traders from acquiring proprietary DOB data would decrease.[2492] As a result, it might be harder for traders who use proprietary DOB feeds (both self-aggregators and traders who use market data aggregators) to generate profits and the competition between those traders would increase. For traders who use core data, the Commission believes that the competition between those traders will increase because the final amendments will reduce the latency and expand the information included in core data, which will allow those traders to devise better trading strategies with bigger profit potential. The Commission believes that the most substantial change in competition will occur between traders who use proprietary DOB feeds (both self-aggregators and traders who use market data

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2490 In this context the term traders could refer to either proprietary traders executing orders on their own behalf or broker-dealers executing orders on behalf of their customers.

2491 Traders who currently subscribe to proprietary DOB feeds may also subscribe to the exclusive SIPs as part of their backup systems. However, the Commission believes that these traders primarily rely on proprietary DOB feeds when making trading decisions because proprietary DOB feeds contain more information and have lower latency than the exclusive SIPs. For additional details and discussion about methods of market data access, see supra Section V.B.2(c).

2492 See supra Section V.C.4(a).

aggregators) and traders who use core data. As described, the final rule expands the information and reduces the latency of core data, thereby closing the gap between core data and proprietary DOB feeds. This will allow traders who use core data to compete on a more level playing field with traders who use proprietary DOB feeds. This will lead to a transfer of profits from traders who use proprietary DOB feeds to traders who use consolidated market data.

Seventh, the Commission believes that the rule changes will affect competition between off-exchange trading venues and exchanges in the market for trading services. As discussed above, the Commission believes that the amendments will reduce the latency of core data.[2493] This could improve the competitive positions of some off-exchange trading venues in the market for trading services. Off-exchange trading venues that currently rely on the exclusive SIPs to calculate the NBBO will benefit from the latency reductions in the distribution of core data provided by the competing consolidators.[2494] These venues will now receive a more timely view of the NBBO, which could improve the execution quality of trades that take place on these venues. This could make them more attractive venues to trade on and they could attract more order flow, from both exchanges and other off-exchange venues. Off-exchange trading venues that currently subscribe to proprietary data feeds could also see their competitive positions improve. If the new core data represents an alternative to the proprietary data feeds for their order executions, they could substitute core data for proprietary data, which could lower their costs. Off-exchange trading venues might be able to pass along these cost reductions as reduced fees to subscribers, which could improve their competitive position relative to exchanges and

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2493 See supra Section V.C.2(c).

2494 Id.

other off-exchange trading venues. Reductions in the fees charged by these off-exchange trading venues could in turn potentially benefit investors if broker-dealers who subscribe to these venues passed along these cost savings by, in turn, reducing their fees.[2495]

3. Capital Formation

The Commission believes the final rule will have a modest impact on capital formation. However, the Commission is unable to quantify the effects on capital formation because, as discussed above, it is unable to quantify the additional gains from trade and the effects of improvements in order routing that may be realized from the rule.[2496] However, in the section below the Commission provides a qualitative description of the effects it believes the rule will have on capital formation.

As discussed above, the Commission believes that the addition of information about odd-lot quotes, depth of book, and auction information to core data may result in more voluntary trades occurring between market participants, which could lead to more efficient gains from trade.[2497] Improved gains from trade may result in a more efficient allocation of capital, which would improve capital formation.

Additionally, the Commission believes that the final amendments will improve order execution for market participants who currently rely upon SIP data, which may lower their

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2495 Broker-dealer subscribers could potentially pass along the cost savings from the reduction in off-exchange trading venue fees to investors either directly, if they reduced fees for investors who were clients of the broker-dealer, or indirectly, if they reduced fees for institutional clients, such as mutual funds, who, in turn, passed along the cost savings to their end investors.

2496 See supra Sections V.C.1(b), V.C.1(c), V.D.1.

2497 See supra Section V.D.1.

transaction costs.[2498] Lower transaction costs could reduce firms' cost of raising capital.[2499] This, in turn could improve capital formation.

E. Alternatives

The Commission considered potential alternatives to the adopted rules that broadly fall into two categories: introduce the decentralized consolidation model and make alternative changes to the core data definition, and maintain the new core data definition in the adopted rules and consider alternative models of SIP competition.

1. Introduce Decentralized Consolidation Model with Addition of Full Depth of Book to Core Data Definition

The Commission considered an alternative that would introduce the decentralized consolidation model and expand core data more than the adopted rules to include information on quotations and aggregate size at all prices in the limit order book ("full depth of book"), including information on aggregated odd-lot sizes at each depth of book level, instead of the depth of book information contained in the adopted rule, i.e., five round lot price levels from the NBBO.[2500] Under this alternative, the definition of a round lot would remain the same as in the adopted rules, which means the costs and benefits associated with the changes in the round lot definition, including changes in the NBBO would be similar to the adopted rule.[2501]

Relative to the adopted rule, full depth of book information would provide market participants who currently do not access proprietary DOB feeds, as well as market participants

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2498 See supra Sections V.C.1(b), V.C.1(c), V.D.1.

2499 See Yakov Amihud and Haim Mendelson, *Asset Pricing and the Bid - Ask Spread*, 17 J. FIN. ECON. 223 (1986).

2500 See supra Section II.F.1.

2501 See supra Section II.D.